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1D Convolutional Autoencoder for ECG Signal Denoising and Classification

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ABSTRACT

Cardiovascular diseases remain a primary global health challenge, requiring dependable electrocardiogram monitoring for early medical intervention. However, ambient clinical noise frequently degrades signal quality, complicating automated diagnostic processing. This paper presents a computationally efficient deep learning framework for joint ECG signal denoising and diagnostic classification. The proposed solution features a lightweight 1D CAE enhanced with SC to preserve crucial morphological traits like the QRS complex during feature extraction. By utilizing an encoder-decoder setup with an internal structural bottleneck, the architecture eliminates severe BW and MA artifacts down to 0 dB conditions. Following the enhancement stage, a dedicated classification head processes the high-fidelity features to categorize heartbeats into specific diagnostic classes. Evaluated on the MIT-BIH Arrhythmia Database, our framework successfully isolates artifacts—achieving an RMSE of 0.0335 under light baseline distortion—while maintaining a classification accuracy of 97.54% and a macro-average F1-score of 92.13%. These outcomes confirm that a highly streamlined convolutional design can deliver robust clinical denoising and reliable arrhythmia detection, making it highly viable for resource-constrained wearable healthcare devices.

INDEX TERMS CAE, Classification, CNN, CVD, Decoder, Denoising, DWT, ECG, EMD, Encoder, MLII, SWT

I. INTRODUCTION

Cardiovascular disease (CVD) is one of the leading causes of death globally, and it involves all disorders that affect the heart and blood circulation or vessels [1]. For this reason, it is necessary to enhance all detection techniques to be able to early detect any irregularities to avoid fatal stages.

One of the most important techniques is the electrocardiogram (ECG), which is a test that can be used to monitor a patient's heart rhythm and electrical activity. The electrical signals produced by the heart are measured using electrodes attached to the patient's skin. The heart activity is recorded to be analyzed by specialists to investigate symptoms of possible heart problems or any abnormalities.

Despite the usefulness of ECG, the recorded signals are prone to various noise sources, which may distort the heart's signal, leading to misdiagnosis. There are many reasons for having noise during measurement.

The following are the main causes of noise:

- Noises caused by electrode movement (EM), which usually have a large amplitude.
- High-Frequency Noises caused by muscle artifact (MA), which is a muscle noise caused by involuntary muscle movements like shivering, tension, or tremors.
- Low-Frequency Noises caused by baseline wander (BW), mainly caused by human breathing /respiration [2].

The mentioned noise types can mask important diagnostic features in the signal, so effectively removing these noises without altering the essential morphology of the ECG signal remains a significant and ongoing challenge[3].

Thus, various denoising techniques have been developed to enhance the quality of ECG signals. This study aims to implement and train a neural network-based filtering/denoising technique using a convolutional autoencoder(CAE).

This paper is structured as follows:

- **Section II** which covers the necessary background information, starting from defining the ECG signal and its characteristics, such as the QRS complex, and waves within ECG cycle like P-waves, and T-waves, then moves to a brief about classical ECG signal denoising techniques, and then introduces the necessity of data-driven techniques, finally it illustrates the needed technologies in this paper like 1D Convolution and Autoencoder.
- **Section III, IV** which outlines the dataset specifications and details the two-stage preparation pipeline for both denoising and heartbeat classification tasks. Furthermore, it comprehensively describes the proposed structural architectures, detailing the layers of the 1D Convolutional Autoencoder (CAE) and the downstream diagnostic classifier.
- **Section VI** which discusses ablation studies comparing different model configurations, tuning parameters, and classification classes to highlight the system's baseline improvements.
- **Section V** which presents the experimental setups, training details, and evaluation metrics used to measure the efficacy of both networks.
- Lastly, **Section VII** explains the major challenges and research gaps, showing that denoising the ECG signal is a very active topic in research, and there are multiple future directions that need to be explored.

II. BACKGROUND AND LITERATURE REVIEW

A. ECG SIGNAL AND ITS CHARACTERISTICS

A full ECG is obtained via 10 electrodes, that captures 12 leads (signals). Each lead identifies the electrical activity from a different angle/perspective in order to get a complete picture of the heart. However, a single lead can offer adequate information for a quick and initial assessment of the patient (Modified Limb Lead 2 (MLII)) is used in this paper, which provides a clear QRS complex and visible P-Waves making it suitable for heartbeat detection). One ECG cycle consists of five waves: P, Q, R, S, and T that map different phases of the heart's activities.

The P-wave is the first positive deflection on the ECG, The QRS represents the simultaneous activation of the right and left ventricles, although most of the QRS waveform is derived from the larger left ventricular musculature and the T-wave should be concordant with the QRS complex. P-R interval measured from the beginning of the P-wave to the first deflection of the QRS complex and Q-T interval measured from the first deflection of QRS complex to end of T-wave at the isoelectric line as shown in Fig. 1 [4].

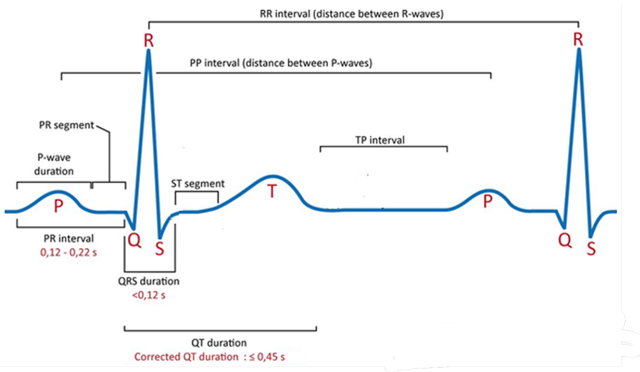


FIGURE 1. ECG Waveform [4]

The nominal values for the mentioned intervals in Lead II are as shown in Table 1.

TABLE 1. Characteristics of normal ECG waves[4]

Waves and intervals	Duration	Amplitude
P-waves	< 0.11 s	< 2.5 mm
QRS-complex	< 0.12 s	> 0.5 mV
T-waves	N/A	N/A
P-R interval	0.12–0.22 s	N/A
S-T interval	N/A	> 0.5 mm
Q-T interval	< 0.45 s	N/A

B. TRADITIONAL ECG SIGNAL DENOISING TECHNIQUES

There are several techniques that addressed ECG signal denoising, including **time-frequency analysis techniques** such as wavelet transforms like discrete wavelet transform(DWT), and Stationary wavelet transform (SWT), and techniques like **empirical mode decomposition (EMD)** and **filtering-based techniques**, which are widely used due to their effectiveness in ECG denoising [5]. These techniques have shown significant results in separating noise from the ECG signal. For instance, wavelet transform-based techniques are efficient at isolating components in both time and frequency domains to remove baseline drift or high-frequency noise. On the other hand, adaptive filters can flexibly adjust to variable noise conditions, while EMD decomposes signals into intrinsic mode functions (IMF) to isolate noise-dominant components [6], [7]. Despite this, such classical techniques are not sufficient as they face notable challenges and limitations that struggle with the critical morphological characteristics of the ECG signal. For instance, muscle artifact noise often overlaps with the frequency bands of P-waves, T-waves, and QRS complexes (important features of ECG signals), leading to signal distortion or loss of clinically relevant information. These overlaps make it difficult for customized filters to distinguish between noise and meaningful signal content, especially under low SNR conditions [8], [9].

C. DATA-DRIVEN ECG SIGNAL DENOISING TECHNIQUES

Traditional ECG denoising techniques, such as Butterworth filters, FIR filters, and Wavelet-based thresholding, have long served as the clinical standard. While these methods have demonstrated effectiveness in basic signal processing [5], they are inherently *rule-based* and rely on predefined mathematical assumptions. This reliance presents several critical limitations that motivate the shift toward data-driven solutions:

- **Manual Parameter Dependency:** As noted by [5], traditional signal processing algorithms have inborn limits in handling various noise types because the selection of many parameters relies too heavily on human expertise and empirical trial-and-error.
- **Morphological Distortion:** Classical linear filters often struggle to differentiate between high-frequency noise and the clinically significant features of the QRS complex. This often leads to "over-smoothing," where vital morphological information is attenuated alongside the noise [7].
- **Static Assumptions:** Methods like Discrete Wavelet Transform (DWT) and Stationary Wavelet Transform (SWT) rely on the choice of wavelet basis functions [5]. If the chosen basis does not perfectly match the signal's underlying characteristics, the denoising performance significantly degrades.

By contrast, data-driven frameworks, such as the proposed Convolutional Autoencoder, learn non-linear mappings directly from the data. Rather than requiring manual parameter tuning or pre-selected basis functions, these models autonomously characterize and suppress stochastic artifacts, offering a robust, adaptive alternative to static rule-based counterparts. Motivation to NN based techniques

D. 1-D CONVOLUTION

1-D Convolution is a fundamental signal processing algorithm in deep learning that is widely used with ECG signals. 1D-CNNs are designed to handle one-dimensional data, such as time-series data, sequences like text or any data where the primary structure is along a single axis. The kernel (or filter) in a 1-D CNN moves along one dimension. If the data is represented as a vector $[x1, x2, ..., xn]$, the kernel will slide over this vector to detect patterns within the sequence. The shape of the kernel is a 1-D array with dimension (k,), where k is the size of the kernel. The kernel slides along the single axis of the input vector. For example, in a time-series application, the kernel moves along the time axis to capture temporal patterns.The receptive fields of a 1-D CNN kernel is a contiguous segment of the input sequence. Therefore, as the kernel slides over the input, it aggregates information from k consecutive elements.For an input sequence x and a kernel w, the convolution operation in a 1D-CNN layer is given as:

$$(x * w)(t) = \sum_{i=0}^{k-1} x(t+i) \cdot w(i)$$

where:

- x is the input sequence,
- w is the kernel (or filter),
- $(x*w)(t)$ denotes the convolution of x and w at position t
- k is the size of the kernel,
- $x(t+i)$ is the element of the input sequence at position $t+i$,
- $w(i)$ is the element of the kernel at position i.

Fig.2 shows how 1-D Convolution operates on the ECG signal.

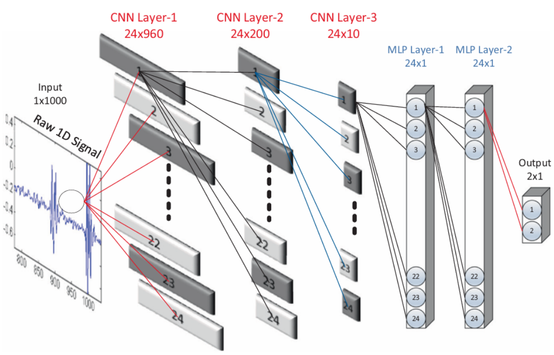


FIGURE 2. 1-D CNN Sample [10]

E. AUTOENCODER

The fundamental architecture of the proposed model is structured into three primary components: the encoder, the bottleneck, and the decoder. The primary objective of the encoder is to take a raw, noisy ECG signal and compress it into a smaller, dense representation. During this phase, the network learns the optimal weights and filters required to map any new incoming signal into this compressed state.

The most critical component of this process is the bottleneck (or latent space). Because the bottleneck restricts the size of the data, the network is forced to learn only the most important, repeating patterns of the cardiac signal, while naturally discarding random, unstructured noise. The size of this bottleneck is a crucial hyperparameter; adjusting its dimensions directly influences the network's learning capacity and dictates how aggressively the model performs the denoising process. Once the noise is filtered out in the bottleneck, the decoder takes this clean, compressed representation and reconstructs it back into the original signal size.

1) Mitigating Signal Blurring

A common challenge during the decoding phase is that reconstructing a large signal from a highly compressed bottleneck can cause "blurring" or over-smoothing. This blurring can distort the sharp, critical features of an ECG, such as the QRS-complex. To resolve this problem, the architecture incorporates Skip Connections. By passing high-resolution features directly from the encoder layers to the corresponding decoder layers, the model successfully recovers sharp morphological details that would otherwise be lost during upsampling. Additionally, regularization techniques such as Dropout are integrated into the network to prevent the model from overfitting to the training noise.

2) Mathematical Formulation

$$h = f(\tilde{x}) = \sigma(W_e * \tilde{x} + b_e)$$

where:

- h is the latent representation (bottleneck),
- $\tilde{x}$ is the noise-corrupted input signal,
- W_e and b_e are the encoder convolutional weights and bias,
- σ is the non-linear activation function,
- $*$ denotes the one-dimensional convolution operation.

$$\hat{x} = g(h) = \sigma(W_d * h + b_d)$$

where:

- $\hat{x}$ is the reconstructed, clean output signal,
- W_d and b_d are the decoder convolutional weights and bias.

$$L_{MSE} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2$$

where:

- L_{MSE} is the Mean Squared Error loss function,
- N is the total number of samples in the batch,
- x_i is the ground-truth clean signal for sample i,
- $\hat{x}_i$ is the reconstructed output signal for sample i.

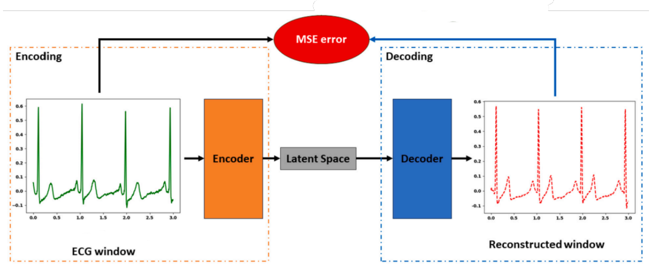


FIGURE 3. Autoencoder Architecture [11]

III. METHODOLOGY

A. DATASET DESCRIPTION

The experimental data utilized in this study is sourced from the MIT-BIH Arrhythmia Database, which serves as the foundational clean ECG record set [12]. The database includes recordings from 47 subjects: 22 women aged 32 to 90 years and 25 men aged 32 to 90 years. Each record has a duration of approximately 30 minutes, digitized at a sampling frequency of 360 Hz with an 11-bit resolution over a 10 mV range. Among these subjects, 25 exhibit normal cardiac rhythms, while 23 present with various arrhythmias. For the purpose of this study, all signals were specifically extracted from Lead II to ensure consistency in morphological analysis.

To evaluate the robustness of the denoising framework, we incorporated noise profiles from the MIT-BIH Noise Stress Test Database [13]. The noise types considered include Baseline Wander (BW), which mimics respiratory-induced baseline shifts; Muscle Artifact (MA), representing electromyographic noise from patient tremors; and Electrode Motion (EM) artifacts. The latter is particularly significant, as EM artifacts often mimic the morphological characteristics of legitimate heartbeats, posing a rigorous challenge to the model's ability to differentiate between physiological events and non-stationary stochastic artifacts.

B. DATA PREPARATION

This study employs a two-stage data preparation pipeline to support both the denoising and classification tasks.

1) Denoising

To create the training data for the denoising model, clean MIT-BIH signals were randomly combined with either Baseline Wander (BW) or Muscle Artifact (MA) noise.

- **Noise Injection:** The noise was injected at three distinct Signal-to-Noise Ratio (SNR) levels: 0, 1.25, and 5. Each segment was corrupted with only one noise type at a time to simplify the learning objective.
- **Segmentation:** The 30-minute recordings were sliced into fixed-length windows of 2048 points, matching the input layer of our proposed network.
- **Normalization:** To ensure numerical stability, all signal amplitudes were normalized to a range of [-1, 1].

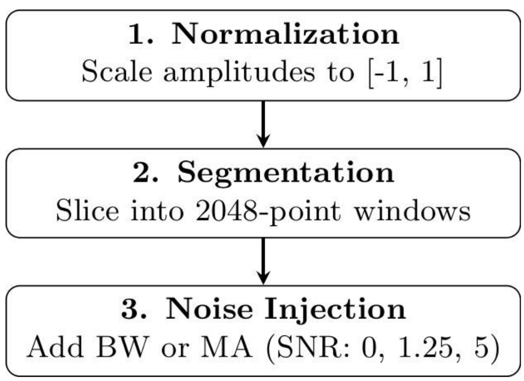


FIGURE 4. Data preparation pipeline for the denoising model.

2) Classification

For the preparation of the dataset for arrhythmia classification, the focus was on heartbeat-level characteristics.

- **Beat Extraction:** Using the annotated R-peak locations, we isolated individual heartbeats by extracting a fixed-length window of 360 samples (180 points before and 180 points after the R-peak).
- **AAMI Standard Mapping:** The raw labels were remapped according to the AAMI standard into three diagnostic classes: Normal (N), Supraventricular (S), and Ventricular (V). Beats categorized as Fusion (F) or Unknown (Q) were excluded.
- **Contextual Input:** To provide the model with rhythmic context, we concatenated the previous, current, and subsequent beats into a single (1, 1080) tensor input.

3) Data Splitting for Denoising

The ECG patient records (45 out of 48) were only used, 3 records were excluded to ensure consistency as they don't have MLII lead channel, which was used across all record. The records were divided into training, validation, and testing sets to ensure intra-patient data so as to avoid data leakage.

MIT-BIH Patient Records Distribution:

- **Training** (35 patients, 78%): 100, 101, 106, 107, 108, 109, 111, 112, 113, 115, 117, 118, 122, 123, 124, 200, 201, 203, 205, 207, 208, 209, 210, 212, 213, 214, 215, 220, 222, 223, 228, 231, 232, 233, 234.
- **Validation** (5 patients, 11%): 105, 103, 119, 217, 221.
- **Testing** (5 patients, 11%): 116, 121, 202, 219, 230.
- **Excluded** (3 patients): 102, 104, 114

Table 2 shows the intra-patients segments distribution among the different data splits, which clearly shows that data is splitted almost equally among noise types and SNR levels in each data split.

TABLE 2. Dataset distribution across training, validation, and testing sets for BW and MA noise types at different SNR levels.

	Training			Validation			Testing		
Noise	**0 dB**	**1.25**	**5 dB**	**0 dB**	**1.25**	**5 dB**	**0 dB**	**1.25**	**5 dB**
BW	3667	3720	3637	535	526	526	507	522	514
MA	3751	3641	3739	518	537	499	548	547	527
Subtotal	22155			3165			3165		

4) Data Splitting for Classification

Table 3 shows the dataset distribution among the three classes N, S, and V, in each split. It clearly shows the class imbalance; which will be handled via FocalLoss discussed in IV-C.

TABLE 3. Distribution of ECG Signal Classes Across Datasets

Class Label	Training	Validation	Testing
N	69,517	8,424	10,554
S	2,703	2	64
V	5,949	1,042	194
Total	**78,169**	**9,468**	**10,812**

IV. ARCHITECTURE

A. OVERALL ARCHITECTURE

The overall system architecture is shown in Fig. 5 which describes how to process the slightly more than 30 min ECG signal (650000 samples) by the designed convolutional Autoencoder (CAE).

The ECG signal is very long, so it is segmented using a sliding window. Based on previous works [14], the segment is of length 2048 samples with 50% overlapping, to ensure that more than 5 cardiac cycles are processed by CAE to provide sufficient information. Moreover, the 50% overlapping is to ensure smooth, continuous transition between the segments, avoiding the loss of crucial information like QRS complexes.

The input to CAE is now 2048 samples segment with either of the noise types (BW or MA) added randomly as discussed before, with SNR Levels $(0db, 1.25db, 5db)$ as suggested by many researches [5], [15], [16], [17] to be able to compare the results with.

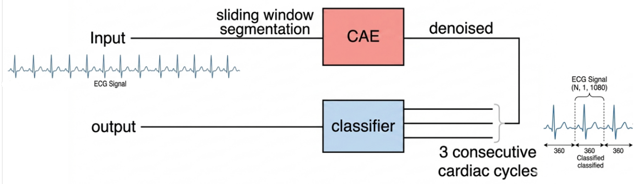


FIGURE 5. Overall System Architecture

More details about the CAE architecture will be discussed in a later section IV-B.

The output of CAE is a denoised segment with equal length as the input segment, which is later reconstructed to make a complete, full denoised ECG signal that is passed afterwards to another type of segmentation to be prepared for classification, which works as follows: first, R peaks are determined in signal, the signal is divided into fragments of 360 samples in length with R peak at the center. It must be noted that 3 consecutive cardiac cycles $(3 \times 360 = 1080$ samples$)$ are input to the classifier with the purpose of classification of the middle segment; this is beneficial for the middle segment to obtain relevant information about the signal from both sides (before and after). The classifier then outputs the label of the middle segment, which can be one of the following labels as illustrated in Table 4:

TABLE 4. Classifier output labels and mapping status based on AAMI standards.

Label	Beat Category	Status
N	Normal / Non-Ectopic Beats	Classified
S	Supraventricular Ectopic Beats (SVEB)	Classified
V	Ventricular Ectopic Beats (VEB)	Classified
F	Fusion Beats	Ignored (Scarcity)
Q	Unknown / Unclassified Beats	Ignored (Scarcity)

B. CAE ARCHITECTURE

The detailed architecture of CAE is shown in Fig. 6, which was inspired by the architecture in research [5].

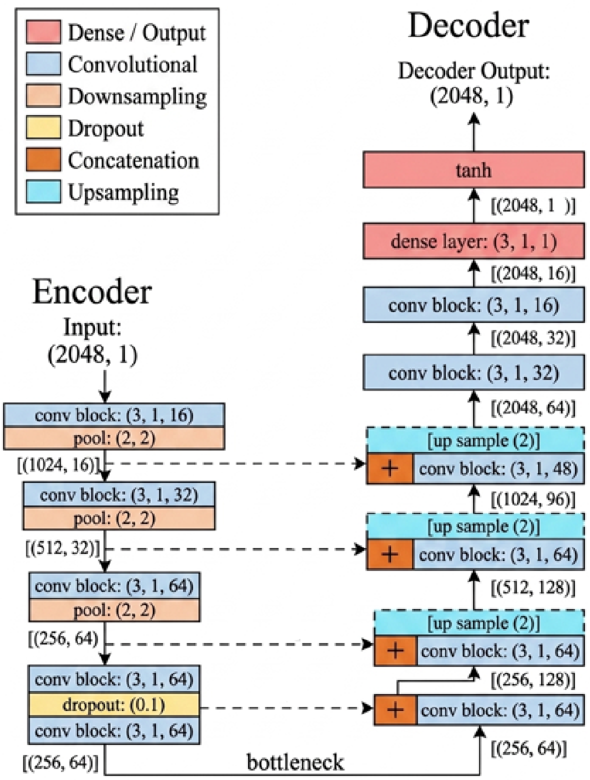


FIGURE 6. Detailed 1D Convolutional Autoencoder Architecture

It is much simplified as it doesn't contain the CNN-SWT module or the transformer as shown in Fig.7, instead the CNN-SWT module is replaced with one of the implementations of Conv. block for more feature extraction as shown in Fig. 8, and based on experimentation, the standard block was the best out of the three.

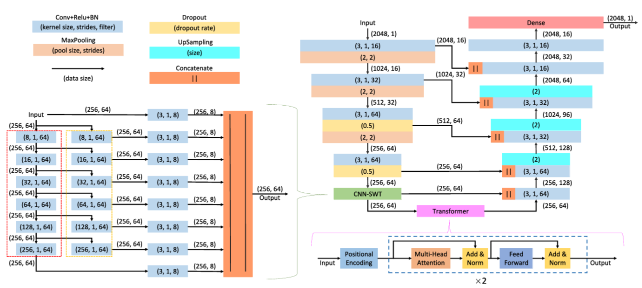


FIGURE 7. The architecture of the denoising module [5].

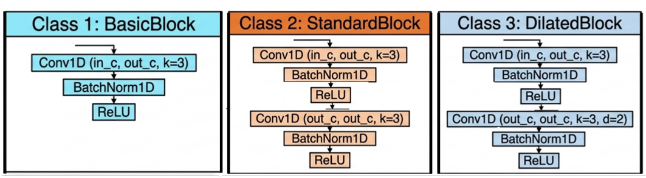


FIGURE 8. Convolution Block Implementations

Denoising Loss Functions

Two denoising loss functions were implemented to be used in the training of 1-D CAE. In section VI will see the results of each during the training.

1) Peak-Weighted Loss ($\mathcal{L}_{\text{PW}}$)

Peak weighted loss function is simply MSE loss function that finds the mean square error between the denoised and clean signal with a coefficient C_P for retaining the peaks during denoising.

$$C_p = 1 + abs(X_c - median(X_c))$$

$$\mathcal{L}_{\text{pw}}(X_c, X_d) = \frac{\sum\left((X_c - X_d)^t \cdot (X_c - X_d) \cdot C_p\right)}{N}$$

where:

- X_c is clean signal, X_d is the denoised signal.
- C_p is the peak coefficient of the clean signal X_c.
- $\mathcal{L}_{\text{pw}}(X_c, X_d)$ is the peak weighted loss fucntion.

2. ECG Combined Loss ($\mathcal{L}_{\text{Total}}$)

The total optimization loss function is a multi-component loss regularized by the following weights w_{mse} , w_{peak} , w_{dist} , and w_{freq} :

$$\mathcal{L}_{\text{Total}} = w_{\text{mse}}\,\mathcal{L}_{\text{MSE}} + w_{\text{peak}}\,\mathcal{L}_{\text{PW}} + w_{\text{dist}}\,\mathcal{L}_{\text{Dist}} + w_{\text{freq}}\,\mathcal{L}_{\text{Freq}}$$

where this is a breakdown of each component of ($\mathcal{L}_{\text{Total}}$) :

- **Mean Squared Error ($\mathcal{L}_{\text{MSE}}$):**

$$\mathcal{L}_{\text{MSE}}(X_c, X_d) = \frac{\sum_{i=1}^{N}\sum_{j=1}^{C}\sum(X_d - X_c)^t(X_d - X_c)}{N}$$

- **Distortion / Derivative Loss ($\mathcal{L}_{\text{Dist}}$):** Preserves ECG morphology and high-frequency slopes using first-order finite differences:

$$\Delta X_k = X_{k+1} - X_k \quad \text{for } k \in [1, T-1]$$

$$\mathcal{L}_{\text{Dist}}(X_c, X_d) = \frac{\sum \Delta X_{d,k} - \Delta X_{c.k}|}{N}$$

- **Frequency Domain Loss ($\mathcal{L}_{\text{Freq}}$):** Ensures spectral consistency using the absolute magnitudes of the real FFT (rFFT):

$$\mathcal{L}_{\text{Freq}}(X_c, X_d) = \frac{\sum \left| |\text{rFFT}(X_c)_f| - |\text{rFFT}(X_d)_f| \right|}{N}$$

C. CLASSIFIER ARCHITECTURE

The detailed architecture of the classifier is shown in Fig. 9 which shows the pipeline for classifying ECG beats, as mentioned before, it takes three consecutive cardiac beats as an input, and classifies the middle one.

The architecture simply consists of the pretrained encoder of the 1-D CAE described in the previous section, combined with the classification head. The pre-trained encoder of 1-D CAE extracts the latent space features that are passed via global average pooling to reduce its temporal size that is flattened to be input to the classification head, which is a sequential fully connected network that outputs the predicition scores of each of 3 classes (N,S,V).

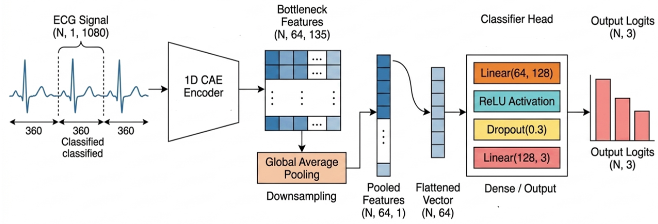


FIGURE 9. Detailed ECG beat classifier Architecture

Focal Loss

To handle extreme class imbalance in the dataset, *FocalLoss* is used, which is an extension of cross-entropy loss, which addresses class imbalance by dynamically down-weighting the loss assigned to easy-to-classify examples during training.
The loss is calculated as follows:
The cross entropy function internally applies a Softmax function to the logits ($-\infty$ to $+\infty$) to transform them into probabilities (0 to 1).
It then extracts the probability belonging specifically to the true target class (p_t). Finally, it computes cross-entropy loss.
Steps of the computations:

$$p_t = \text{Softmax}(\mathbf{z}_{\text{target}}) = \frac{e^{z_{\text{target}}}}{\sum_j e^{z_j}}$$

$$\mathcal{L}_{\text{CE}} = -\log(p_t)$$

Where $\mathbf{z}$ represents the raw vector of input logits, and z_{target} denotes the logit value corresponding to the ground-truth label.
The cross-entropy loss $\mathcal{L}_{\text{CE}}$ is scaled using a focusing parameter γ and class-balancing weight vector α_t to give the focal loss $\mathcal{L}_{\text{Focal}}$:

$$\mathcal{L}_{\text{Focal}} = \alpha_t(1-p_t)^{\gamma} \cdot \mathcal{L}_{\text{CE}}$$

The final output is the average loss across the entire batch: $\bar{\mathcal{L}} = \frac{1}{N}\sum \mathcal{L}_{\text{Focal}}$.

V. EVALUATION AND RESULTS

A. ECG DENOISING EVALUATION METRICS

For objective assessment of the performance of CAE on ECG signal denoising, 3 metrics are utilized:Mean Squared Error (MSE), Percent Root-Mean-Square Difference (PRD), and SNR Improvement (SNR_{Imp}).

1) **Mean Squared Error (MSE) :**

$$\text{MSE}(X_c, X_d) = \sum (X_c - X_d)^2$$

2) **SNR Improvement (SNR_{Imp})** The signal-to-noise ratio improvement quantifies the explicit amount of noise power eliminated by the autoencoder architecture:

$$\text{SNR}_{\text{Imp}} = \text{SNR}_{\text{denoised}} - \text{SNR}_{\text{noisy}}$$

where the **Signal-to-Noise Ratio (SNR)** evaluates the ratio of the desired signal power to the residual noise power, expressed in decibels (dB).

- **Denoised SNR ($\text{SNR}_{\text{denoised}}$):** Computed using the clean signal and the reconstructed signal, where the residual noise is defined as $n = x - \hat{x}$:

$$\text{SNR}_{\text{denoised}} = 10 \cdot \log_{10}\left(\frac{power(X_c)}{power(X_c - X_d)}\right)$$

- **Noisy SNR ($\text{SNR}_{\text{noisy}}$):** Computed using the clean signal and the raw noisy input signal:

$$\text{SNR}_{\text{noisy}} = 10 \cdot \log_{10}\left(\frac{power(X_c)}{power(X_c - X_n)}\right)$$

3) **Percent Root-Mean-Square Difference (PRD)** which evaluates the morphological degradation and distortion level between the target signal and reconstructed/ denoised output as a percentage value:

$$\text{PRD}(X_c, X_d) = \sqrt{\left(\frac{power(X_c - X_d)}{power(X_c)}\right)} \times 100\%$$

B. CAE DENOISING RESULTS

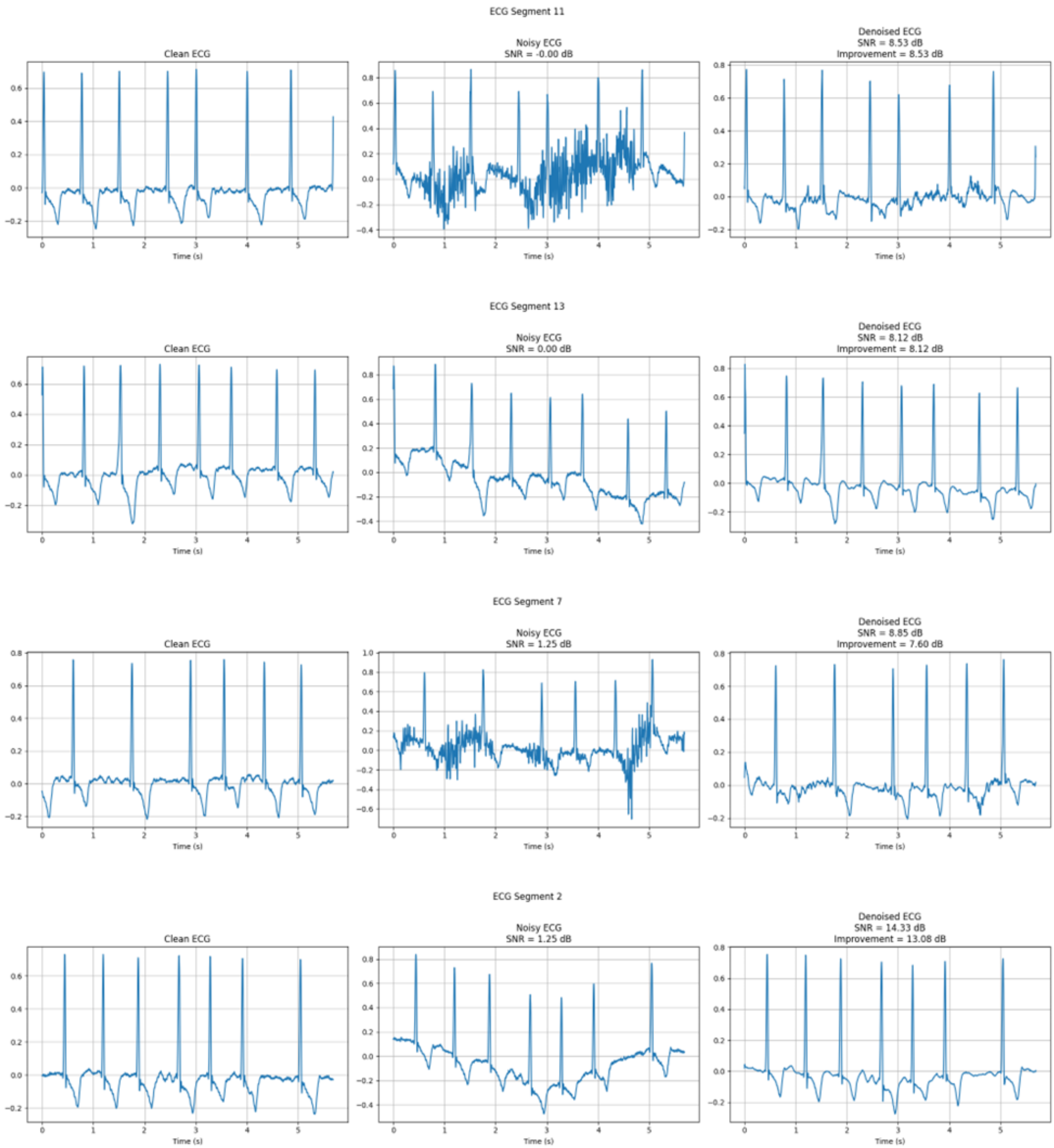
Table 5 shows the results of $ResStackCAE_{\text{Micro}}$with a smaller bottleneck than the original $ResStackCAE$ using $\mathcal{L}_{\text{CE}}$, which will be later be analyzed to demonstrate the effect of bottleneck sizes and denoising loss functions.
Many things can be interpreted from the table such as:

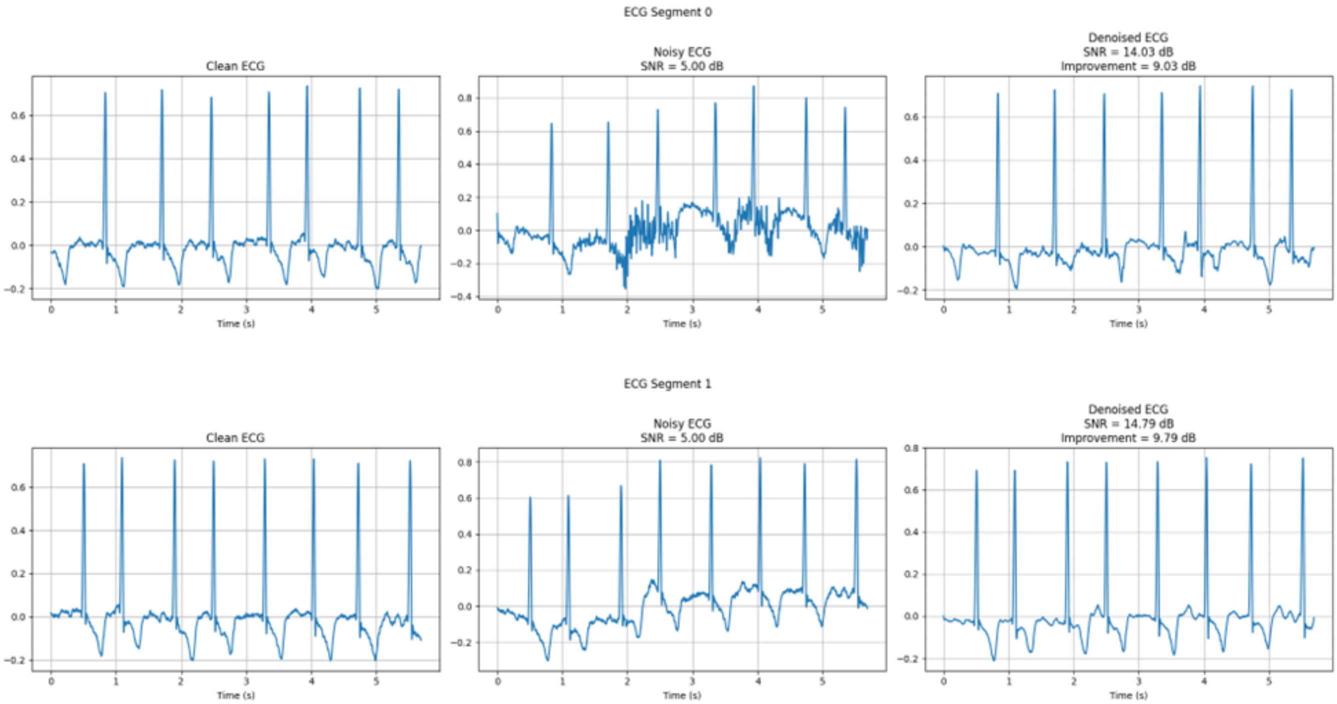
1) The CAE has a slightly better denoising effect on signals with added BW noise type than signals with added MA noise type.
2) Higher SNR_{improv} is achieved with more noisy signals (lower SNR in dB), which makes sense since the signal is noisier, there is more room for improvement.
3) Lower values of PRD after denoising indicate lower distortion between the clean signal and the denoised or reconstructed signal.
4) All these values are the result of training the CAE (very simple architecture in comparison with denoising module of CNN-trans[5]) for 5 epochs, which shows significant improvement for the denoising process despite the models simplicity.

The following waveforms compare the clean, noisy, and denoised segments to show the effect of denoising visually.

TABLE 5. ECG Denoising Performance Evaluation Across Noise Types and SNR Levels

Noise Type	SNR Level	SNR Improv.	Denoised SNR	Noisy PRD	Denoised PRD	Noisy MSE	Denoised MSE
BW	0.00	8.11	8.11	100.00	41.46	0.01567	0.00459
	1.25	7.54	8.79	86.60	38.70	0.01027	0.00288
	5.00	5.04	10.04	56.23	34.31	0.00418	0.00220
MA	0.00	7.25	7.25	100.00	45.68	0.01407	0.00401
	1.25	6.94	8.19	86.60	41.18	0.01117	0.00352
	5.00	4.67	9.67	56.23	35.31	0.00450	0.00255





C. ECG CLASSIFICATION EVALUATION METRICS

For an objective assessment of the performance of ECG classifier, multiple metrics are utilized

1) Overall Accuracy

Accuracy measures the percentage of correctly classified instances across the entire test dataset containing N total samples:

$$\text{Accuracy} = \frac{\sum_{i=1}^{C} TP_i}{N}$$

Notice, due to class imbalance, this metric doesn't hold great significant meaning, but an increased value is +ve indication, but not enough to judge the classifier.

2) Per-Class and Macro-Averaged Metrics

Because the dataset exhibits a severe class imbalance, macro-averaging is utilized to assign equal weight to each classification category regardless of its sample frequency.

- **Macro Precision:** Measures the average precision (predictive correctness) for all classes:

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i}$$

$$\text{Precision}_{\text{macro}} = \frac{1}{C}\sum_{i=1}^{C} \text{Precision}_i$$

- **Sensitivity (Recall):** Measures the model's capability to detect all true positive instances. Both the per-class sensitivity and its macro-averaged counterpart are computed as:

$$\text{Sensitivity}_i = \frac{TP_i}{TP_i + FN_i}$$

$$\text{Sensitivity}_{\text{macro}} = \frac{1}{C}\sum_{i=1}^{C} \text{Sensitivity}_i$$

- **F_1-Score:** Computes the harmonic mean of precision and sensitivity. Both the individual class scopes and global macro-averages are established:

$$F_{1,i} = 2 \cdot \frac{\text{Precision}_i \cdot \text{Sensitivity}_i}{\text{Precision}_i + \text{Sensitivity}_i}$$

$$F_{1,\text{macro}} = \frac{1}{C}\sum_{i=1}^{C} F_{1,i}$$

D. ECG CLASSIFIER RESULTS

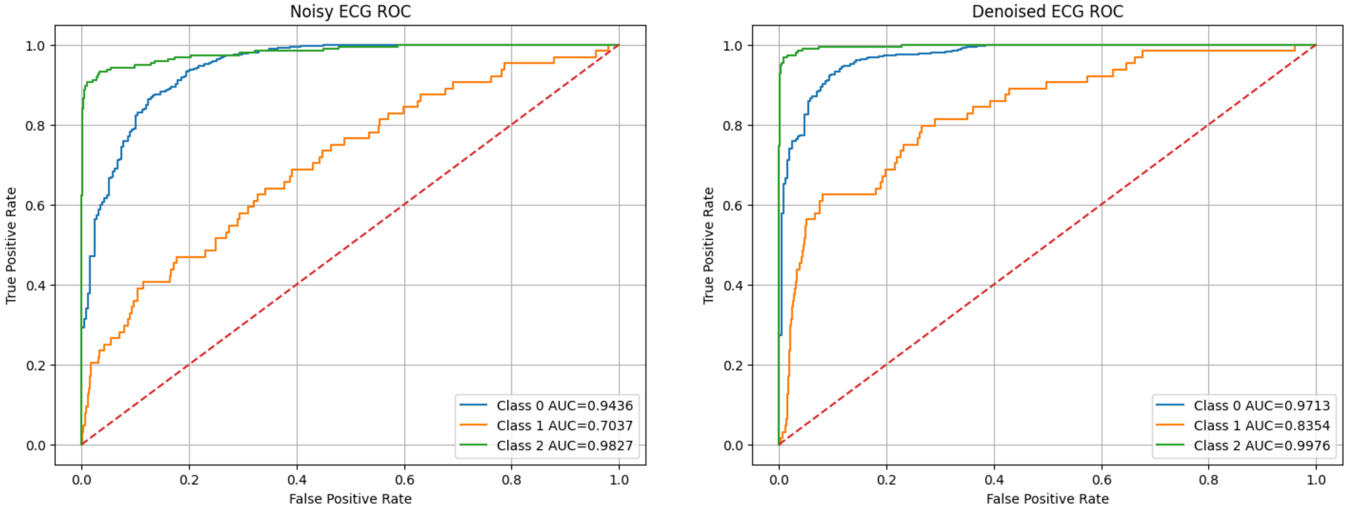
Classification Metrics Comparison Between Noisy and Denoised ECG Signals

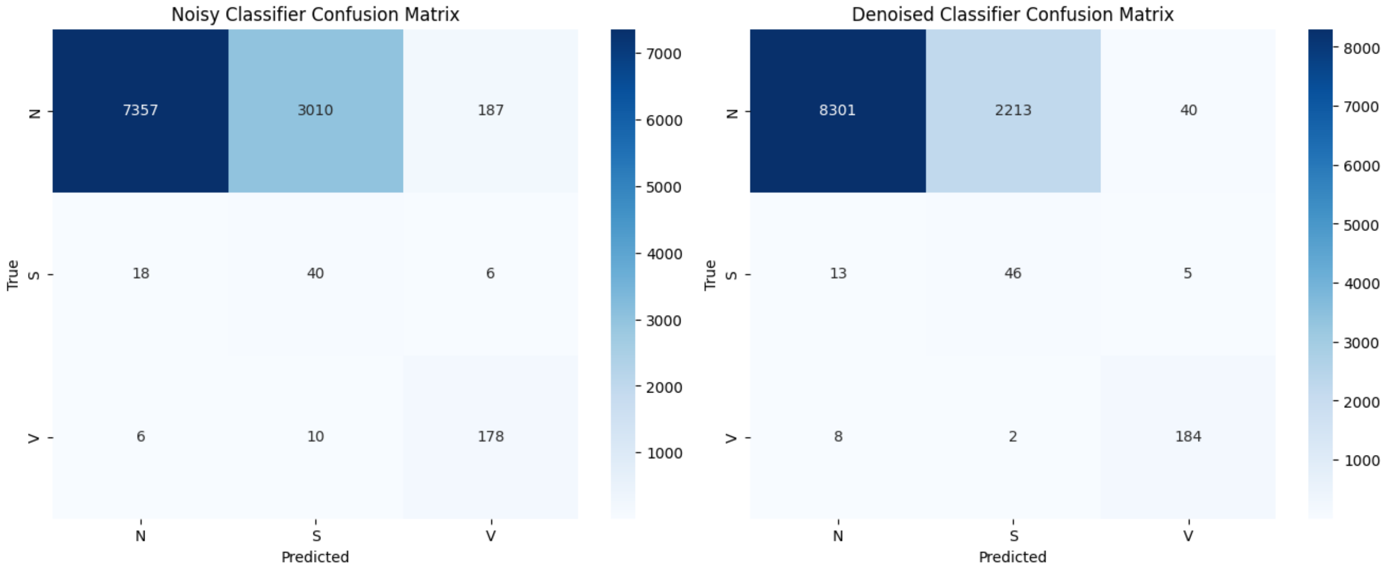
TABLE 6. Classification Metrics Comparison for Noisy and Denoised ECG Signals

Metric	Noisy ECG	Denoised ECG
Accuracy	0.7006	0.7890
Precision	0.4965	0.6071
Sensitivity	0.7465	0.8179
F1-score	0.4920	0.5964
Mean AUC	0.8767	0.9348

TABLE 7. Per-Class Performance Comparison of F1-Score, Sensitivity, and AUC

Class	F1 Noisy	F1 Denoised	Sensitivity Noisy	Sensitivity Denoised	AUC Noisy	AUC Denoised
N	0.8196	0.8646	0.6958	0.7624	0.9482	0.9722
S	0.0260	0.0398	0.6094	0.7969	0.6844	0.8304
V	0.5136	0.8304	0.9278	0.9588	0.9827	0.9973





VI. ABLATION STUDIES

A. DENOISING ABLATION

TABLE 8. Ablation Performance Comparison of ResStackCAE with different bottlenecks using the $\mathcal{L}_{\text{total}}$

Model Variant	Test Loss	Noisy SNR (dB)	Denoised SNR (dB)	SNR Improv. (dB)
ResStackCAE	0.03986	2.07	8.66	6.59
ResStackCAE_Micro	0.04021	2.07		**6.89**
ResStackCAE_Macro	0.05067	2.07	8.70	6.63

Table 8 shows that the models that gives best denoising results is ResStackCAE_Micro using $\mathcal{L}_{\text{total}}$

TABLE 9. Denoising Performance Direct Comparison of Loss Functions Across Identical Bottleneck Variants

Model Variant	Loss Function	Test Loss	Noisy SNR (dB)	Denoised SNR (dB)	SNR Improv. (dB)
ResStackCAE (Base)	$\mathcal{L}_{\text{PW}}$ (Peak-Weighted)	0.00365	2.07	8.47	6.40
	$\mathcal{L}_{\textbf{Total}}$ **(Combined)**	0.03986	2.07	**8.66**	**6.59**
ResStackCAE_Micro	$\mathcal{L}_{\text{PW}}$ (Peak-Weighted)	0.00358	2.07	8.59	6.52
	$\mathcal{L}_{\textbf{Total}}$ **(Combined)**	0.04021	2.07	**8.96**	**6.89**
ResStackCAE_Macro	$\mathcal{L}_{\textbf{PW}}$ **(Peak-Weighted)**	0.00348	2.07	**8.82**	**6.75**
	$\mathcal{L}_{\text{Total}}$ (Combined)	0.05067	2.07	8.70	6.63

Table 9 shows that the base and Micro bottleneck of ResStackCAE gives better denoising results with $\mathcal{L}_{\text{total}}$, while Macro bottleneck gives better denoising results with $\mathcal{L}_{\text{PW}}$

B. CLASSIFICATION ABLATION

TABLE 10. Downstream ECG Classification Ablation Analysis: Performance Impact of the Proposed Denoising Pipeline

Metric	Class	Noisy ECG	Denoised ECG	Improvement
Global Accuracy	–	0.7006	0.7890	+0.0884
F_1**-Score**	Class N	0.8204	0.8795	+0.0591
	Class S	0.0256	0.0396	+0.0140
	Class V	0.6301	0.8700	+0.2399
Sensitivity (Recall)	Class S	0.6250	0.7188	+0.0938
	Class V	0.9175	0.9485	+0.0309
ROC-AUC	Class N	0.9436	0.9713	+0.0278
	Class S	0.7037	0.8354	+0.1316
	Class V	0.9827	0.9976	+0.0149
Correct Predictions (Count)	Class N	7,357	8,301	+944
	Class S	40	46	+6
	Class V	178	184	+6

Table 10 shows that the convolutional autoencoder improved the downstream ECG classification performance. This indicates that removing noise helped the classifier learn more discriminative cardiac features despite the classifer was trained for just 3 epochs.

VII. CHALLENGES AND FUTURE DIRECTIONS

Several constraints within the current framework offer opportunities for future study. First, model validation relies entirely on single-lead recordings from Lead II configurations. Clinical environments typically utilize multi-lead monitoring arrangements, such as standard 12-lead setups, to track spatial electrical properties of cardiac anomalies. Adjusting the 1D CAE pipeline to handle multi-channel inputs simultaneously is therefore a vital progression toward identifying complex arrhythmias.

Second, the system must be evaluated across diverse patient populations and distinct hardware systems. Because different institutional databases contain specialized baseline distortion properties and varying digitization frequencies, future work should prioritize cross-database validation to establish generalization without retraining. Additionally, while the system effectively counteracts severe BW and MA interference down to 0 dB, real-world monitoring introduces irregular, dynamic noise profiles such as electrode motion.

Finally, future investigations will explore the integration of lightweight attention architectures or transformer units within the structural bottleneck. This development aims to capture extensive chronological dependencies while keeping the model parameters low. Maintaining this computational efficiency will ultimately support the migration of the combined framework to resource-constrained hardware configurations, including ambulatory diagnostic sensors and point-of-care medical wearables.

VIII. CONCLUSION

In this study, we developed an integrated framework for the enhancement and diagnostic classification of ECG signals. The process began with meticulous data preparation, wherein clean signals were partitioned into fixed-length segments and augmented with synthetic artifacts—specifically BW and MA—to simulate realistic clinical noise conditions.

The core of our proposed solution consists of a dual-stage architecture. First, a 1D CAE was employed for signal denoising. This model utilizes an encoder to compress noisy input signals into a condensed latent representation known as the bottleneck. During the subsequent decoding phase, we integrated dropout layers and SC to effectively reconstruct the signal. These architectural choices were critical in preserving high-fidelity cardiac morphologies and preventing signal blurring. Our comparative analysis demonstrated that the 1D CAE with Skip Connections significantly outperformed traditional filtering methods (such as DWT, EMD, and SWT). Under 5 dB BW noise, the model achieved a superior RMSE of 0.0335 and a PRD of 20.93%, while also demonstrating robust recovery against complex MA noise with an RMSE of 0.0594 at the 5 dB level.

Finally, the denoised signals were passed through a dedicated classification architecture to determine the patient's cardiac condition. By effectively cleaning the input data before classification, our model ensures that the diagnostic head operates on high-quality, noise-free features, achieving an overall classification accuracy of 97.54% and a macro-average F1-score of 92.13% under evaluation conditions. This end-to-end approach demonstrates that a robust 1D CAE, when coupled with a precise classification stage, offers a reliable and efficient solution for automated arrhythmia detection.

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